

# Mykhaylo M. Malakhov

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## Education

### University of Minnesota Twin Cities

Minneapolis, MN

*PhD in Biostatistics*

09/2020 – 07/2025

- Committee: Wei Pan (advisor), Eric F. Lock, Thierry Chekouo, Arslan A. Zaidi
- Dissertation: “Accounting for Interactions and Uncertainty in Multi-Omic Association Studies”

### Andrews University

Berrien Springs, MI

*BS in Mathematics (minor in Computing)*

08/2016 – 05/2020

- *Summa Cum Laude* and J. N. Andrews Honors Scholar
- Honors thesis: “Managing White-nose Syndrome in Bats: A Spatially Dynamic Modelling Approach”

### Budapest Semesters in Mathematics

Budapest, Hungary

*Study abroad*

09/2019 – 12/2019

- Selective study abroad program focusing on advanced mathematics

## Experience

### Research

#### Stanford University School of Medicine

Stanford, CA

*Postdoctoral Scholar, Department of Epidemiology and Population Health*

08/2025 – present

- Developing personalized biomarkers for precision disease screening by integrating genomics, proteomics, and health history through machine learning
- Quantifying the heritability and genetic architecture of second primary lung cancer
- Mapping genetic variants that regulate chimeric gene fusion transcripts in tumors

#### University of Minnesota School of Public Health

Minneapolis, MN

*Graduate Research Assistant, Division of Biostatistics and Health Data Science*

09/2023 – 07/2025

- Introduced the co-expression-wide association study (COWAS) method for identifying genes and proteins whose genetically regulated co-expression is associated with complex traits
- Explored the impact of aging on the genetic regulation of protein expression and co-expression, as well as the downstream effects of age-related dysregulation on Alzheimer’s disease
- Contributed to the development of Mendelian randomization methods for causal inference with multiple correlated exposures (MVMR-cML-SuSiE) and multiple related outcomes (MR2-SuSiE)
- Contributed to a study showing that imputation of Alzheimer’s disease can enhance the performance of nonlinear TWAS and PWAS

#### Denali Therapeutics

South San Francisco, CA

*Intern, Human Genetics*

06/2023 – 08/2023

- Built cloud-based software that enabled scientists across the company to easily clean, store, and access genome-wide association study (GWAS) data
- Leveraged public GWAS data to better understand the molecular mechanisms of genetic variants associated with neurodegenerative diseases

#### University of Minnesota School of Public Health

Minneapolis, MN

*Predocotrinal Trainee, Division of Biostatistics*

09/2020 – 05/2023

- Was funded by a T32 Training Grant from the National Institutes of Health (NIH)
- Proposed and implemented the differential regulation analysis by bootstrapping (DRAB) framework, a statistical approach for identifying genes with context-specific patterns of local genetic regulation
- Contributed to the first study demonstrating nonlinear effects of gene expression on complex traits

## **UCLA Institute for Pure and Applied Mathematics**

*Researcher and Project Manager, RIPS program*

**Los Angeles, CA**

*06/2019 – 08/2019*

- Worked on the Air Force Research Laboratory team in the Research in Industrial Projects for Students (RIPS) program
- Proposed novel attractor reconstruction and model calibration methods
- Inferred reaction rate coefficients for hydrogen-oxygen combustion from a time series of one observable

## **Williams College**

*Student Researcher, SMALL REU program*

**Williamstown, MA**

*06/2018 – 08/2018*

- Project 1: demonstrated how to improve management outcomes for white-nose syndrome in bats by considering metapopulation dynamics
- Project 2: established guidelines for transboundary infectious disease management when multiple administrative jurisdictions set different objectives

## **Andrews University**

*Research Assistant, Department of Mathematics*

**Berrien Springs, MI**

*06/2017 – 08/2017*

- Modeled the effects of climate change on seabird behavior and population dynamics
- Proved that egg cannibalism and egg-laying synchrony can yield strong Allee effects, which allow gull colonies to survive at higher sea surface temperatures than otherwise possible

## **Teaching**

### **Andrews University**

*Teaching Assistant, Department of Mathematics*

**Berrien Springs, MI**

*01/2017 – 04/2020*

- Tutored students of all levels in the Mathematics Center
- Led a short course on LaTeX
- Graded for Foundations of Advanced Mathematics
- Served as a substitute lecturer for Calculus I and II

## **Honors and awards**

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### **National and international**

- Roger Williams Memorial Award, International Genetic Epidemiology Society (IGES) (11/2024)
- Reviewers' Choice Abstract Award, American Society of Human Genetics (ASHG) (11/2024)
- Goldwater Scholarship, Barry Goldwater Scholarship & Excellence in Education Foundation (03/2018)

### **Stanford University**

- CEHG Postdoctoral Fellowship, Center for Computational, Evolutionary and Human Genomics (2026 – 2027)

### **University of Minnesota**

- Audience Favorite Student Poster Award, Division of Biostatistics and Health Data Science (05/2025)
- Student Member of the Delta Omega Honorary Society in Public Health, Pi Chapter (05/2025)
- People's Choice Poster Award, School of Public Health Research Day (04/2022)
- 2nd Place Best Poster Award, School of Public Health Research Day (04/2022)
- 3rd Place, University of Minnesota Interdisciplinary Health Data Competition (04/2022)
- Dean's PhD Scholars Award (09/2020)

### **Andrews University**

- Harold T. Jones Scholarship for highest excellence in mathematics (04/2018)
- Louis Ulloth Scholarship for most significant leadership contributions (04/2018)
- Member of Pi Mu Epsilon, Michigan Gamma Chapter (03/2018)
- Highest score in the Putnam Competition among students at Andrews University (2018, 2019)
- Excellence awards for 12 mathematics courses (2016 – 2020)
- Dean's List (awarded every semester 2016 – 2020)
- Four-year full tuition ACT/SAT scholarship (2016 – 2020)

## Publications

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The \* symbol denotes equal contributions.

### Working papers and preprints

1. **Malakhov, M. M.** & Pan, W. Age-dependent regulation of protein expression and its impact on Alzheimer's disease. In preparation.
2. **Malakhov, M. M.** & Henson, S. M. Backward bifurcations in a low-dimensional model of seabird population dynamics. In preparation.
3. Chan, L. S., **Malakhov, M. M.** & Pan, W. A multi-response Mendelian randomization framework for identifying shared and distinct causes of related traits. In preparation.

### Peer-reviewed journal articles

4. **Malakhov, M. M.** & Pan, W. Co-expression-wide association studies link genetically regulated interactions with complex traits. *Nature Communications* **16**, 11061. <https://doi.org/10.1038/s41467-025-66039-6> (Dec. 2025).
5. He\*, R., Ren\*, J., **Malakhov, M. M.** & Pan, W. Enhancing nonlinear transcriptome- and proteome-wide association studies via trait imputation with applications to Alzheimer's disease. *PLOS Genetics* **21**, e1011659. <https://doi.org/10.1371/journal.pgen.1011659> (Apr. 2025).
6. Chan, L. S., **Malakhov, M. M.** & Pan, W. A novel multivariable Mendelian randomization framework to disentangle highly correlated exposures with application to metabolomics. *The American Journal of Human Genetics* **111**, 1834–1847. <https://doi.org/10.1016/j.ajhg.2024.07.007> (Sept. 2024).
7. **Malakhov, M. M.**, Dai, B., Shen, X. T. & Pan, W. A bootstrap model comparison test for identifying genes with context-specific patterns of genetic regulation. *The Annals of Applied Statistics* **18**, 1840–1857. <https://doi.org/10.1214/23-AOAS1859> (Sept. 2024).
8. Lin, Z., Xue, H., **Malakhov, M. M.**, Knutson, K. A. & Pan, W. Accounting for nonlinear effects of gene expression identifies additional associated genes in transcriptome-wide association studies. *Human Molecular Genetics* **31**, 2462–2470. <https://doi.org/10.1093/hmg/ddac015> (July 2022).
9. Blackwood, J. C., **Malakhov, M. M.**, Duan\*, J., Pellett\*, J. J., Phadke\*, I. S., Lenhart, S., Sims, C. & Shea, K. Governance structure affects transboundary disease management under alternative objectives. *BMC Public Health* **21**, 1782. <https://doi.org/10.1186/s12889-021-11797-3> (Oct. 2021).
10. Duan\*, J., **Malakhov\*, M. M.**, Pellett\*, J. J., Phadke\*, I. S., Barber, J. & Blackwood, J. C. Management efficacy in a metapopulation model of white-nose syndrome. *Natural Resource Modeling* **34**, e12304. <https://doi.org/10.1111/nrm.12304> (Aug. 2021).

### Theses and technical reports

11. **Malakhov, M. M.** *Accounting for Interactions and Uncertainty in Multi-Omic Association Studies*. PhD Dissertation (University of Minnesota, July 2025). <https://hdl.handle.net/11299/277379>.
12. **Malakhov, M. M.**, Fitzpatrick, B. R., Lopez, R. A. & Shivkumar, A. *Attractor Reconstruction and Empirical Parameter Inference for Hydrogen-Oxygen Chemistry*. Technical Report AD1098889 (Air Force Research Laboratory, Aug. 2019). <https://apps.dtic.mil/sti/citations/AD1098889>.
13. **Malakhov, M. M.** *Managing White-nose Syndrome in Bats: A Spatially Dynamic Modelling Approach*. Honors Thesis (Andrews University, Apr. 2019). <https://doi.org/10.32597/honors/216>.

## Presentations

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### Conference talks

1. “Genetic Dysregulation of Protein Expression in Aging and Neurodegeneration.” Platform Session on Omics; International Genetic Epidemiology Society (IGES) Annual Meeting; Cologne, Germany. (Sept. 2025)

2. "Co-expression-wide association studies link genetically regulated interactions with complex traits." Contributed Session on Genetics; STATGEN Conference; Minneapolis, MN. (May 2025)
3. "Co-expression-Wide Association Studies Implicate Protein-Protein Interactions in Complex Disease Risk." Session on Leveraging Transfer Learning and AI for Complex Data and Predictions; ENAR Spring Meeting; New Orleans, LA. (Mar. 2025)
4. "Co-expression-Wide Association Studies Implicate Protein-Protein Interactions in Complex Disease Risk." Williams Award Platform Presentation Session; International Genetic Epidemiology Society (IGES) Annual Meeting; Denver, CO. (Nov. 2024)
5. "A bootstrap model comparison test for identifying genes with context-specific patterns of genetic regulation." Session on Mediation and Interaction in Genomics; Joint Statistical Meetings (JSM); Toronto, ON, Canada. (Aug. 2023)
6. "Accounting for nonlinear effects of gene expression in transcriptome-wide association studies." Andrews Research Conference (ARC); Andrews University; Berrien Springs, MI. (May 2022)
7. "Modeling the impact of bat dispersal on white-nose syndrome control strategies." Mathematics Section; Michigan Academy of Science, Arts, and Letters (MASAL); Alma College; Alma, MI. (Mar. 2019)
8. "Federalism in Epidemic Modeling: Multi-objective Management of Interconnected Populations." AMS-MAA-SIAM Special Session on Research in Mathematics by Undergraduates and Students in Post-Baccalaureate Programs; Joint Mathematics Meetings (JMM); Baltimore, MD. Jointly with Ishan Phadke. (Jan. 2019)
9. "Cannibalism and synchrony in a periodic matrix seabird population model." Mathematics Section; Michigan Academy of Science, Arts, and Letters (MASAL); Central Michigan University; Mount Pleasant, MI. (Mar. 2018)
10. "Backward Bifurcations in a Periodic Matrix Model of Seabird Population Dynamics." MAA General Contributed Paper Session on Modeling and Applications; Joint Mathematics Meetings (JMM); San Diego, CA. (Jan. 2018)

### **Seminar and symposium talks.....**

1. "Accounting for interactions and uncertainty in statistical genetics." Eigen\* Mathematics and Physics Seminar; Andrews University; Berrien Springs, MI. (Feb. 2025)
2. "An introduction to AWK for biostatistics." Division of Biostatistics and Health Data Science Student Seminar; University of Minnesota; Minneapolis, MN. (Dec. 2024)
3. "Identifying genes with tissue-specific patterns of genetic regulation." Genomic Data Science Mini-Symposium; Masonic Institute for the Developing Brain (MIDB); University of Minnesota; Minneapolis, MN. (Oct. 2022)
4. "Attractor Reconstruction and Empirical Parameter Inference for Hydrogen-Oxygen Chemistry." Projects Day; Institute for Pure and Applied Mathematics (IPAM); University of California, Los Angeles; Los Angeles, CA. Jointly with Brianna Fitzpatrick, Rebecca Lopez, and Abhishek Shivkumar. (Aug. 2019)
5. "Application of Convergent Cross Mapping to Chemical Reactions." Edwards Air Force Base; Boron, CA. Jointly with Brianna Fitzpatrick, Rebecca Lopez, and Abhishek Shivkumar. (Aug. 2019)
6. "Managing White-nose Syndrome in Bats: A Spatially Dynamic Modelling Approach." Honors Thesis Symposium; Andrews University; Berrien Springs, MI. (Apr. 2019)

### **Poster presentations.....**

1. "Co-expression-wide association studies link genetically regulated interactions with complex traits." National Graduate Student Symposium; St. Jude Children's Research Hospital; Memphis, TN. (Mar. 2025)
2. "Co-expression-wide association studies implicate protein-protein interactions in complex disease risk." Statistical Genetics and Genetic Epidemiology Poster Session; American Society of Human Genetics (ASHG) Annual Meeting; Denver, CO. (Nov. 2024)
3. "Leveraging Public GWAS Data to Inform Discovery." Summer Research Symposium; Denali Therapeutics; South San Francisco, CA. (Aug. 2023)
4. "Identifying genes with tissue-specific patterns of genetic regulation." School of Public Health Research Day; University of Minnesota; Minneapolis, MN. (Apr. 2023)

5. "Governance structure affects transboundary disease management under alternative objectives." School of Public Health Research Day; University of Minnesota; Minneapolis, MN. (Apr. 2022)
6. "Data-driven Attractor Reconstruction and Parameter Inference for Hydrogen-Oxygen Chemistry." MAA Student Poster Session; Joint Mathematics Meetings (JMM); Denver, CO. (Jan. 2020)
7. "Managing White-nose Syndrome in Bats: A Spatially Dynamic Modeling Approach." Honors Scholars and Undergraduate Research Poster Symposium; Andrews University; Berrien Springs, MI. (Mar. 2019)
8. "Efficacy of Control in a Spatially Dynamic Model of White-nose Syndrome." Summer Science Poster Session; Williams College; Williamstown, MA. Jointly with Ishan Phadke. (Aug. 2018)
9. "A Periodic Matrix Model of Seabird Behavior and Population Dynamics." Honors Scholars and Undergraduate Research Poster Symposium; Andrews University; Berrien Springs, MI. (Mar. 2018)

## Volunteering and academic service

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### Peer reviewer for:

- *The American Journal of Human Genetics* (1)
- *BMC Bioinformatics* (1)
- *Scientific Reports* (1)
- *Biogerontology* (1)
- *Ecology and Evolution* (1)
- *Human Heredity* (1)

### International Genetic Epidemiology Society (IGES)

*Communications and Young Investigators Committee Member*

**Remote**

*01/2025 – present*

I am serving on the Communications Committee and on the Young Investigators Committee of the International Genetic Epidemiology Society (IGES). In these roles, I manage the society's social media presence and help organize events for early career researchers.

### University of Minnesota School of Public Health

*BCOE Member*

**Minneapolis, MN**

*08/2022 – 05/2024*

As a member of the Biostatistics Community Outreach and Engagement Committee (BCOE), I mentored high school students in clinical trial design and data analysis. I also participated in and helped coordinate a wide range of volunteering opportunities for students, faculty, and staff in the Division of Biostatistics and Health Data Science.

### Pi Mu Epsilon: The National Mathematics Honor Society

*President, Michigan Gamma Chapter*

**Berrien Springs, MI**

*08/2018 – 05/2020*

In my role leading the Pi Mu Epsilon chapter at Andrews University, I arranged game nights and social gatherings for math majors, planned an annual Pi Day party, and conducted our chapter's induction ceremonies. After one year as President I was reelected for a second term.

### Engineers Without Borders USA

*Vice President, Andrews University Chapter*

**Berrien Springs, MI**

*08/2018 – 05/2019*

I coordinated outreach activities, assisted with engineering design, and planned travel itineraries for a student-led solar energy project at a remote school in Madagascar. The summer of 2018 I traveled to Madagascar to help conduct the assessment phase of our project.

### Andrews University

*Eigen\* Mathematics President*

**Berrien Springs, MI**

*08/2017 – 05/2018*

As Mathematics President of Eigen\*, the math/physics student organization at Andrews University, I planned seminar series, weekend retreats, and other social and educational events. I also helped organize our university's first preparation course and team for the Putnam Competition, the most prestigious undergraduate mathematics competition in the world.

### Engineers Without Borders USA

*Treasurer, Andrews University Chapter*

**Berrien Springs, MI**

*08/2017 – 05/2018*

I oversaw expenditures, wrote grant applications, and organized fundraising efforts for the Andrews University chapter of Engineers Without Borders. Under my leadership, our chapter raised about \$20,000 for a solar energy project in Madagascar.